

WRRRA M 0 9 Common Sector Accounting and the Upstream Input Contract

Arithmetic calibration on one denominator and connection to particle channels

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Evaluation scope and completed calculation

Version 0.9 computes phenotype, resident nonphenotype and return using one address measure, then connects the expressed budget to the selected 0.8 particle channels. Verification inputs are WRRRA Core 1.0, MCC 2.3.2, frozen 0.8 inputs, and the upstream two-stage filter and zeta-zero update calculations. Known particle assignments and composition targets are legitimate calibration inputs for this constructed model.

The WRRRA-specific transformation takes the address state and spectrum, accumulates admission from a depleted reservoir, closes the three-sector ledger at the recovery boundary, and applies normalized conditional channel routing followed by the computed selection permutation. Outputs are 5%, 26.8% and 68.2%, resident Actual 31.8%, total accounted weight 100%, and a 48-channel ledger. Twenty-seven verification groups pass, and all nine 0.8 results and their charge assignments are reproduced unchanged.

Falsification conditions are failure of frozen calibrated reproduction, negative or incomplete sector effects, inconsistent transition accounting, duplicated channel or generation weight, charge disagreement, or undisclosed calibration changes. Mapping arithmetic load to physical energy and pressure belongs to 0.10; individual 0/1 measurement outcomes and physical records belong to 0.13. Admission weights here are deterministic expected transport, with no newly implemented physical Born measurement rule.

Address state and calibration on a common denominator

Addresses run from 2 through $N_{\text{address}}=1,000,000$. Every sector uses the same $n^{-\alpha}$ measure. Primes initially occupy the SOURCE reservoir, even composites form resident nonphenotype D, and odd composites support phenotype admission. The physical interpretation and origin of this arithmetic classification remain constitutive assumptions and upstream questions.

$$w_n = \frac{n^{-\alpha}}{\sum_{m=2}^{N_{\text{address}}} m^{-\alpha}}, \sum_n w_n = 1. \quad (1)$$

The input $\alpha=1.8996876950554356$ is the arithmetic calibration fitting $D=0.268$. The threshold $h=1.447673170352440$ fits phenotype 0.05. Return 0.682 follows by closure. Beta in the constant-admission comparison and β_{eff} in the zeta update have different roles. In the default update, $\beta_{\text{eff}}=0.8654570124137115$ is computed as phenotype divided by available odd-composite weight and is not supplied as an extra free coefficient.

The 4.8769% comparison admits all odd composites at alpha=2, while 26.7942% is the even-composite comparison at alpha=1.9. Values across those rows are never added. Each row has its own normalized denominator; the joint reference instead uses alpha about 1.899687695 and the update threshold h.

alpha	Odd composite %	Even composite %	Prime %
2.0	4.876893	24.999961	70.123145
1.9	5.774215	26.794200	67.431586

Upstream inputs and the update boundary

Interface wrra.upstream-ledger/1 fixes the address state, measure, adopted spectrum, coefficients and phases, admission update, recovery boundary, channel coupling and source hashes. The reference address state is a normalized diagonal n^α distribution; the computational API also checks supplied normalized diagonal states. Large-scale dynamics of a general full address density matrix are outside this run. Downstream carrier rho and the address state are inputs on different state spaces, coupled by the declared conditional routing rule.

The spectrum is the first four positive zeta-zero heights gamma_j, rechecked by independent high-precision computation and Euler Maclaurin expansion. Each coefficient is 1/2, initial phases are zero, xi=0.1, and there are K=8 admission updates. Selecting zeta heights and cosine driving is an upstream model input. Update index k and xi are dimensionless; seconds and proper time are not assigned. Physical spectral energies require allowed modes and boundary conditions in 0.12.

$$\delta_n(k) = \sum_{j=1}^J c_j \cos[\gamma_j(\log n + \xi k) + \theta_j]. \quad (2)$$

$$a_n(k) = \frac{1}{1 + e^{h - \delta_n(k)}}, T_n(K) = 1 - \prod_{k=0}^{K-1} (1 - a_n(k)). \quad (3)$$

New admission b_n is removed from the still-unadmitted odd-composite reservoir r_n . Adding attempts without depletion would count previously admitted weight again. Cumulative products and sequential depletion are computed independently and agree. Even composites occupy D from the start; primes and unadmitted odd composites remain pending SOURCE S until the recovery boundary.

$$b_n(k) = r_n(k) a_n(k), r_n(k+1) = r_n(k) - b_n(k), r_n(0) = w_n. \quad (4)$$

At the declared boundary $k=K$, pending S is transferred to the return ledger R. Subsequent resident folds use an identity update, the retention rule of this release. Its origin and dynamical cosmological recovery are not derived. S and R are never counted twice. Row $k=-1$ denotes the state before admission and is not a negative physical time.

$$e_\varphi(n) = 1_{oddcomp} T_n(K), e_D(n) = 1_{evencomp}, e_R(n) = 1 - e_\varphi(n) - e_D(n). \quad (5)$$

$$f_s = \sum_n w_n e_s(n), e_s(n) \geq 0, \sum_s e_s(n) = 1, \sum_s f_s = 1. \quad (6)$$

$$\varphi_k + D_k + S_k + R_k = 1, R_{k < K} = 0, S_{k \geq K} = 0. \quad (7)$$

k	Phenotype %	Resident D %	Pending S %	Return R %
-1	0.000000	26.800000	73.200000	0.000000

k	Phenotype %	Resident D %	Pending S %	Return R %
0	1.501794	26.800000	71.698206	0.000000
1	2.513899	26.800000	70.686101	0.000000
2	3.044795	26.800000	70.155205	0.000000
3	3.704795	26.800000	69.495205	0.000000
4	4.050025	26.800000	69.149975	0.000000
5	4.630648	26.800000	68.569352	0.000000
6	4.853503	26.800000	68.346497	0.000000
7	5.000000	26.800000	68.200000	0.000000
8	5.000000	26.800000	0.000000	68.200000
9	5.000000	26.800000	0.000000	68.200000
10	5.000000	26.800000	0.000000	68.200000

Resident Actual and complete Actual scopes

Upstream resident Actual is phi+D after the recovery boundary and occupies 31.8% of the full address denominator. The downstream complete Actual ledger accounts for phenotype, resident nonphenotype and the returned provenance of a background response, totaling 100%. This does not mean returned R remains in the resident address state. Its provenance is represented separately in the downstream response ledger.

$$A_{resident} = f_{\phi} + f_D = 0.318, A_{completeledger} = f_{\phi} + f_D + f_R = 1. \quad (8)$$

$$f_{\phi|A} = \frac{f_{\phi}}{f_{\phi} + f_D}, f_{D|A} = \frac{f_D}{f_{\phi} + f_D}, f_{R|A} = 0. \quad (9)$$

Renormalization within the resident scope gives about 15.72327% phenotype and 84.27673% D. These conditional shares differ from the full 5%, 26.8% and 68.2% partition. The following sector, resident and complete rows all use the original denominator. Stored inputs and ledgers remain unrounded; only displayed tables are rounded.

Scope	Original denominator	Share %
Phenotype phi	1	5.00000000
Resident nonphenotype D	1	26.80000000
Return R	1	68.20000000

Scope	Original denominator	Share %
Resident Actual	1	31.80000000
Complete accounted Actual	1	100.00000000

Conditional connection to selected particle channels

The actual 0.8 selection F_{DX} and its permutation P on sixteen origins are executed unchanged. Phenotype address weight is routed through a conditional origin kernel $q_i(n)$, then the selected permutation and normalized generation state p_g are applied. The reference uses an address-independent 1/16 kernel and 1/3 per generation. A family classified by its odd smallest prime can supply a different normalized kernel; nonuniform kernels and generation weights are executed as controls.

$$x_i = \sum_n w_n e_\varphi(n) q_i(n), \sum_i q_i(n) = 1, y = P x. \quad (10)$$

$$W_{g,i} = p_g y_i, P^\dagger P = I, \sum_g p_g = 1, \sum_{g,i} W_{g,i} = f_\varphi. \quad (11)$$

The address-to-channel correspondence is a disclosed constitutive input. It does not uniquely derive particle identities or masses from prime genealogy. Y and Q use the same 0.8 charge operator. The weights below divide the reference 5% phenotype budget and are not measured particle populations or new cosmic composition fractions. The inventory retains 45 Standard-Model chiral components and three conditional neutral extension slots; their existence, mass and occupation are not measured here.

Field	Count	Y	Full ledger weight %
Q_L	18	1/6	1.87500000
L_L	6	-1/2	0.62500000
u_c	9	-2/3	0.93750000
d_c	9	1/3	0.93750000
nu_c	3	0	0.31250000
e_c	3	1	0.31250000

The next boundary to physical energy and pressure

The common arithmetic partition is 5%, 26.8% and 68.2%, while the 0.6 to 0.8 physical energy calibration remains 4.93%, 26.5% and 68.57%. Separate ledgers preserve both; arithmetic calibration never silently overwrites physical inputs. Existing 0.8 load and pressure calculations leading to gravity and expansion are reproduced. Returned provenance is structurally assigned to the background-response branch in `physical_bridge`, but its energy and pressure functions remain unset.

Version 0.10 must specify physical energy load $\epsilon_s(n,V)$, sector coupling and volume dependence. The following relation is a requirement for that interface, not an executed 0.9 output. The fixed state mathcal S means the address state, effects and coupling configuration, distinct from pending SOURCE S_k. R=68.2% alone does not determine pressure, and 95:5 alone does not determine length. Existing expansion response and the future causal account of twist remain distinct. The SI defining value of c stays a fixed input.

$$E_s(V) = \sum_n w_n e_s(n) \epsilon_s(n, V), P_s = - \left. \frac{\partial E_s}{\partial V} \right|_s. \quad (12)$$

Subsequent stages separate update order from physical time, address cutoff from zeta cutoff and information capacity, and local thresholds from global Capacity. Microscopic generation of upstream inputs, global overload and reset, initial opening and initial cutoff selection remain upstream questions. Software execution provenance is not called a physical observation record or an accumulated twist record.

Verification and reproduction

The 27 groups cover independent primality, frozen upstream code and results, independently checked zeta heights, cumulative products versus depletion, address and frame conservation, Actual denominators, equality of all 0.8 results, nonuniform channel coupling, arbitrary diagonal states and effect positivity on a small coherent state, explicit arithmetic refitting, boundary changes, equal aggregates from different microscopic rules, separate cutoffs, rejection of sixteen invalid inputs, and provenance hashes. Revision r1 also checks twenty exhaustive sparse-kernel cases; direct positive accumulation preserves zero channel support without negative cancellation residuals.

Changing K to 4 and 16 with other calibrations frozen gives phenotype about 3.704795% and 5.682474%. With $\xi=0$, explicitly refitting h restores 5% while address admissions differ. Aggregate reproduction therefore does not establish a unique microscopic rule. Explanatory reproduction under validated calibration and declared rules is accepted; prediction is reserved for later unmeasured outputs after model fixation.

```
python -m pip install -r calculations/wrra_m_0_9/requirements.txt
python calculations/wrra_m_0_9/run_release.py
```

From the archive root, the command generates the input and result JSON, frame, scope, address and channel CSVs, and verification report. Parameters.json is the single configuration input, embedding the 0.8 input and five upstream snapshots. A clean copy removes captured outputs before execution and checks byte-identical reproduction and the SHA256 manifest. Bilingual equations and calculated tables are matched.

Input SHA256 7986e6afcb26e3546111387f34f953147e16cb3c8cb6b900264adf214e51e95

Subsequent development order

The sequence continues with 0.10 energy and pressure mapping; 0.11 sequential calibration $G \rightarrow H_0 \rightarrow f_{\phi} \rightarrow f_c \rightarrow m_e c^2$; 0.12 shutter, proper time and spectra; 0.13 quantization, observation and records; 0.14 cutoffs and capacity; 0.15 stress, twist and size; 0.16 neutrino mass, mixing and oscillation; 0.17 propagation in the same geometry; and 1.0 integration audit and fixation. Each version closes through verification input, WRRRA-specific transformation, output and falsification conditions.

References

Choi Wonsik. WRRRA-M 0.1 to 0.8 and upstream hypotheses. GitHub research repository. <https://github.com/Wonsik-Choi-janefather/wrra-m-0.1>

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Choi Wonsik. Minimal Computation Cosmology 2.3.2. Common carrier baseline. <https://github.com/Wonsik-Choi-janefather/minimal-computing-cosmology-2.3.2>

NIST Digital Library of Mathematical Functions. Sections 25.10 and 25.11 iii. Zeta zeros and Euler Maclaurin representation. <https://dlmf.nist.gov/25.10> <https://dlmf.nist.gov/25.11>